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Project Management 5.3

Project Scheduling

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Learning Outcomes

- By the end of this week, you will be able to:
  - State the importance of scheduling
  - Discuss the differences between different schedule methods
  - Identify the need to schedule
  - Give relevant schedule examples

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Importance of Project Schedules

- Managers often cite delivering projects on time as one of their biggest challenges
- Average time overrun from 1995 CHAOS report was 222%; improved to 163% in 2001 study
- Time has the least amount of flexibility; it passes no matter what
- Schedule issues are the main reason for conflicts on projects, especially during the second half of projects.

Derived from Pinto, J. (2020) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

Project Time Management Processes

- Project time management involves the processes required to ensure timely completion of a project. Processes include:
  - **Activity definition:** Project charter includes start and end dates and budget information, Scope statement and WBS help define what will be done, develop realistic duration estimates
  - **Activity sequencing:** Using this you can draw a Critical Path Analysis (CPA (project network analysis))
  - **Activity duration estimating:** Decide the timing for the project and for the activity in terms of number of days to be allocated and hours for the activity, also the elapsed time. The team should create the estimated activity (days and hours) and the PM should review and approve it.

Derived from Pinto, J. (2020) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

Critical Path Method (CPM)

- CPM is a project network analysis technique used to predict total project duration
- A critical path for a project is the series of activities that determines the *earliest time* by which the project can be completed
- The critical path is the *longest path* through the network diagram and has the least amount of slack or float.

Derived from Pinto, J. (2020) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

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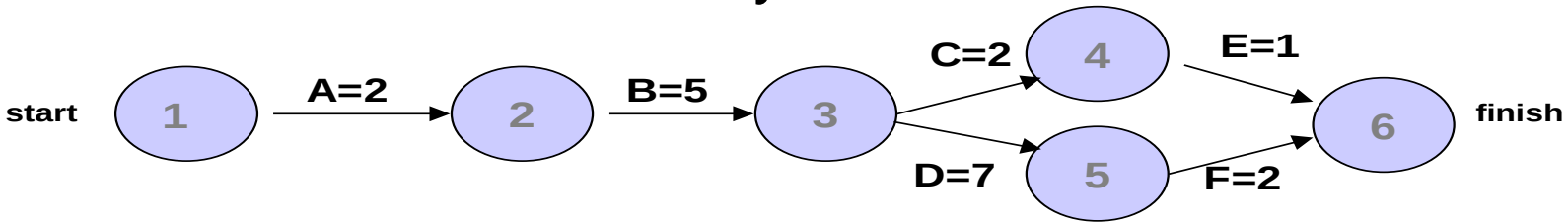
Finding the Critical Path

- First develop a good project network diagram (AOA)
- Add the durations (days) for all activities on each path through the project network diagram
- The longest path is the critical path.

Derived from Pinto, J. (2020) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

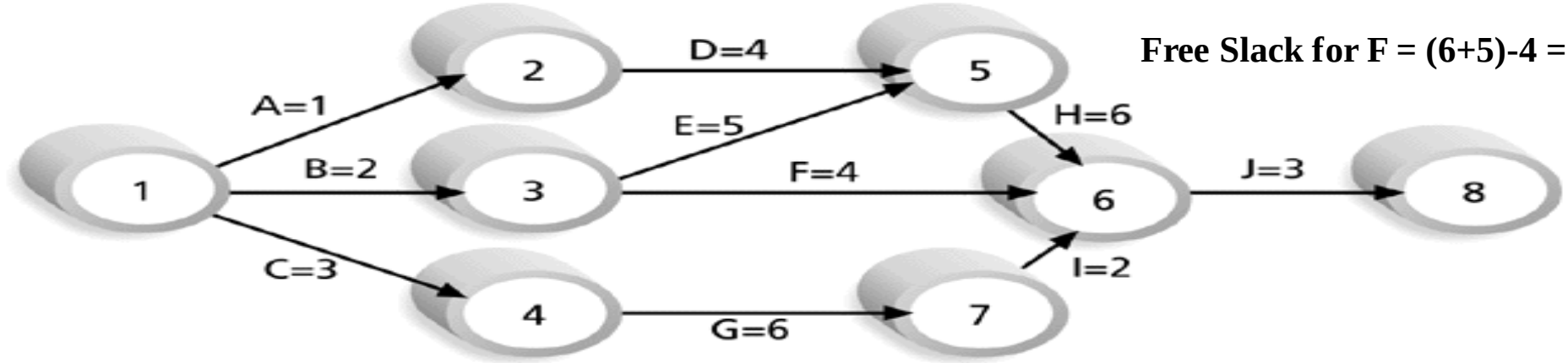
Simple Example of Critical Path Analysis

- Consider the following project network diagram. Assume all times are in days.



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Determining the Critical Path for a Project



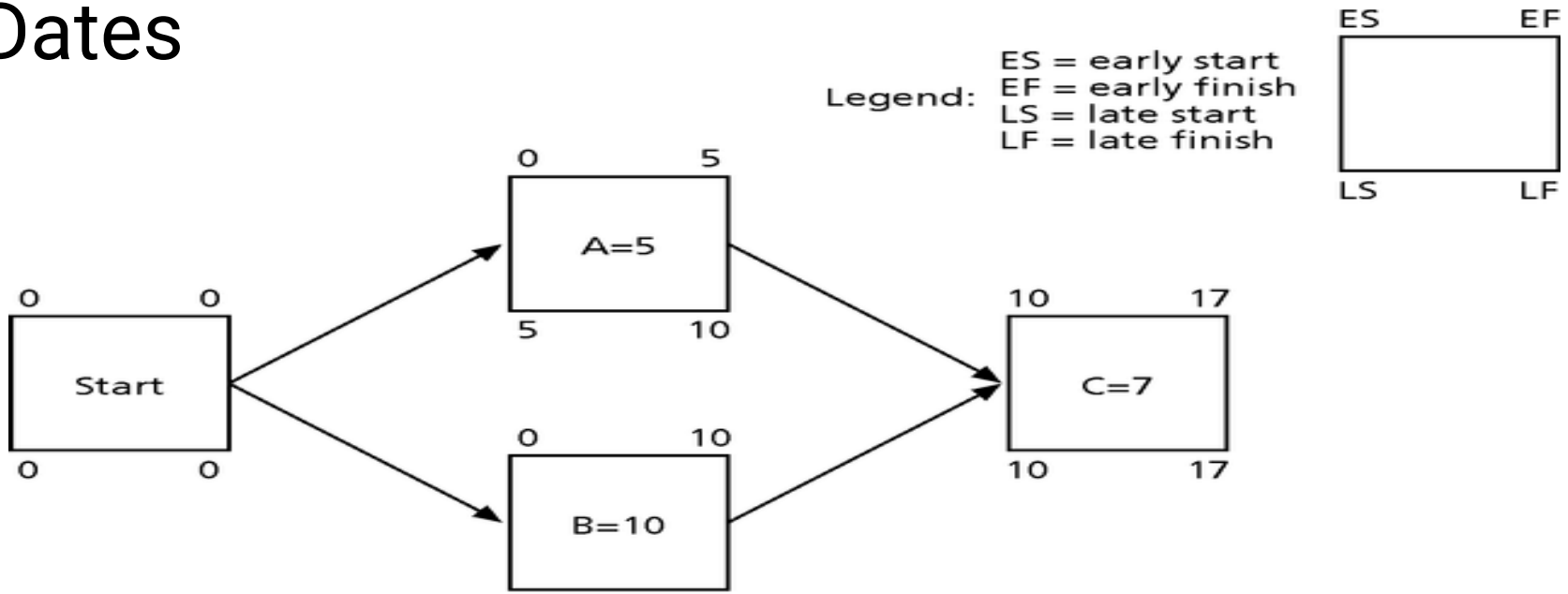
Note: Assume all durations are in days.

Free Slack for F = (6+5)-4 = 7

(2d total slack)
(7d total slack)
(2d total slack)

Derived from Pinto, J. (2020) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

Calculating Early and Late Start and Finish Dates



Derived from Pinto, J. (2020) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

Theory of Constraints & Critical Chain Project Scheduling

A constraint limits any system's output.

The Goal – Dr. Goldratt (1984)

TOC Methodology

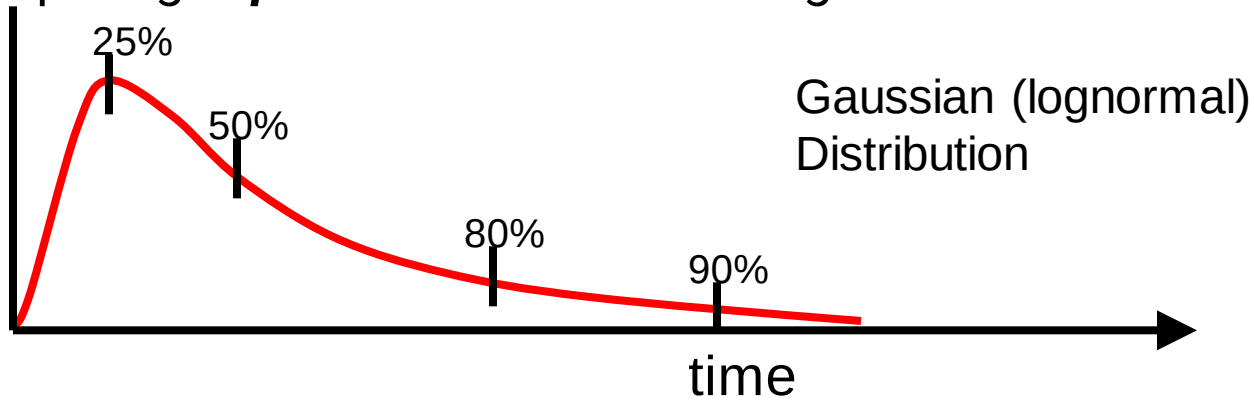
1. Identify the constraint
2. Exploit the constraint
3. Subordinate the system
4. Elevate the constraint
5. Repeat the process

Derived from Pinto, J. (2020) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

Critical Chain PM (CCPM)and the Causes of Project Delay

How safety is added to project activities

1. Individual activities ***overestimated***
2. Project manager ***safety margin***
3. Anticipating ***expected cuts*** from management



Derived from Pinto, J. (2020) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

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CCPM Changes

- Due dates & milestones ***eliminated***
- ***Realistic estimates*** – 50% level not 90%
- ***"No blame"*** culture
- Subcontractor deliveries & work ***scheduled ES***
- ***Non critical*** activities scheduled ***LS***
- Factor the effects of ***resource contention***
- ***Critical chain usually not the critical path***
- Solve resource conflicts with ***minimal disruption***

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CCPM Critiques

- ***No milestones*** used
- ***Not significantly different*** from PERT
- ***Unproven*** at the portfolio level
- ***Anecdotal support*** only
- ***Incomplete*** solution
- Overestimation of activity ***duration padding***
- ***Cultural changes*** unattainable

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Look Forward

- Week six
  - Project Planning Tools
  - Project Governance
  - Project Lifecycles
  - Needs Analysis